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****Software Requirements Specification (SRS)****

****1. Introduction****

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for a hospital management system. This system aims to facilitate the management of patient appointments, doctor schedules, medical records, and reports.

1.2 Scope

This SRS covers the functionalities required for the hospital management system, including user registration, appointment booking, schedule management, patient record maintenance, and report generation.

1.3 Definitions, Acronyms, and Abbreviations

- HMS: Hospital Management System
- UI: User Interface
- API: Application Programming Interface
- SMS: Short Message Service
- DPS: Data Protection Standards

1.4 References

[1] IEEE-standard for SRS-documents, IEEE Std 830-1998

1.5 Overview

This SRS details the requirements for a HMS that enables patients and staff to manage appointments, maintain medical records, generate reports, and adhere to data protection standards.

****2. Overall Description****

2.1 Product Perspective

The HMS is intended for use by hospitals to improve the management of patient appointments, doctor schedules, and medical records. It provides a user-friendly interface for both patients and staff members.

2.2 Product Functions

- User authentication and authorization
- Appointment booking, updating, and canceling
- Doctor schedule management
- Patient record maintenance
- Report generation (daily and monthly)

- Email and SMS appointment reminders
- Admin user privileges for managing users, appointments, and reports

2.3 User Characteristics

The HMS caters to patients, doctors, administrative staff, and system administrators with varying roles and access levels.

2.4 Constraints

- The system must support at least 300 concurrent users (REQ-008).
- All patient data must be securely stored and comply with DPS (REQ-009).

2.5 Assumptions and Dependencies

The HMS assumes the availability of a stable internet connection and email/SMS services for appointment reminders.

3. Functional Requirements

REQ-001: The system allows patients and staff to register and login using their email and password

REQ-002: Patients should be able to book, update, and cancel appointments with doctors

REQ-003: Doctors should be able to view their schedules and manage patient records

REQ-004: The system allows staff to maintain patient medical histories and prescriptions

REQ-005: The system sends appointment reminders via email or SMS

REQ-006: Admin users should be able to manage doctors, patients, and appointments

REQ-007: The system generates daily and monthly reports of hospital activities

4. Non-Functional Requirements

4.1 Performance Requirements

- **The system shall respond to search queries within 2 seconds under normal load (to be defined) [AI-SUGGESTED]**
- The HMS shall support at least 300 concurrent users without response time exceeding 3 seconds

4.2 Security Requirements

- All user data must be encrypted during transmission and storage
- Access to sensitive patient data should be restricted based on user roles

4.3 Usability Requirements

- The system shall provide intuitive, easy-to-use interfaces for all users (patients, doctors, administrative staff, system administrators)
- **Error messages shall be clear and actionable [AI-SUGGESTED]**

4.4 Reliability Requirements

- **The HMS shall have an uptime of at least 99% [AI-SUGGESTED]**
- **The system shall perform regular backups of all data [AI-SUGGESTED]**

4.5 Maintainability Requirements

- **The system shall allow for easy addition, modification, or removal of users and their roles** [AI-SUGGESTED]
- **The system shall provide a logging mechanism for tracking user activity and system errors** [AI-SUGGESTED]

4.6 Portability Requirements

- **The HMS shall be designed to run on commonly used operating systems (e.g., Windows, Linux, macOS)** [AI-SUGGESTED]

5. External Interface Requirements

5.1 User Interfaces

- **The system shall have a web-based user interface (UI) accessible via browsers** [AI-SUGGESTED]
- Mobile application UIs are optional but recommended for improved patient convenience

5.2 Hardware Interfaces

- No specific hardware interfaces are required

5.3 Software Interfaces

- **The HMS shall support various APIs for integration with hospital systems and third-party services (e.g., email, SMS providers)** [AI-SUGGESTED]

5.4 Communication Interfaces

- **The system shall communicate via standard internet protocols (HTTP/S, SMTP) for data transmission** [AI-SUGGESTED]

6. System Constraints

[No specific constraints provided in the requirements]

7. Assumptions and Dependencies

The HMS assumes the availability of a stable internet connection and email/SMS services for appointment reminders. Integration with existing hospital systems is optional but may require additional development efforts.